

MATTHEW YANG

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EDUCATION

University of Waterloo

Bachelor of Computer Science & Finance

Sep 2022 – Apr 2027

GPA: 91%

- **Courses:** Operating Systems, Machine Learning, Compilers, Algorithms, Databases, Probability, Statistics

WORK EXPERIENCE

Cubist Systematic Strategies

May 2026 – Aug 2026

Quantitative Developer Intern | Python, Ray

New York, NY

- Built an **Airflow** pipeline verifying the correctness of in-house intraday factor return calculations against **MSCI Barra**, driving **firm-wide** upstream data quality fixes across **14 factor models**
- Expanded central backtester to support international and multi-country factor models for global equity coverage
- Parallelized historical backfill step of factor model onboarding with **Ray**, cutting runtime from 1 day to **2 hours**

Augment

Jan 2026 – Apr 2026

Software Engineering Intern | Node.js, TypeScript, AWS

San Francisco, CA

- Engineered **"agentic custom tools"**, allowing end-users to write tool call implementations in TypeScript to customize agent behavior, eliminating **10+ hours** of engineering work from new customer onboarding
- Enforced **safe execution of untrusted code** on AWS Lambda via process-level isolation and module allowlisting
- Designed the data model and versioning system for custom tools to enable per-brokerage rollbacks

Point72

Jun 2025 – Aug 2025

Software Engineering Intern | Python, TypeScript, React

New York, NY

- Led the rewrite of a trade reconciliation platform to support a major macro trading system migration, delivering a **UAT-ready system in under 6 weeks**
- Implemented caching layer with async database polling in **Flask API** for **80% reduction** in API latency
- Modernized legacy 1,000-line SQL stored procedures into **pandas**-based Python modules with 90% test coverage

Onex

Jan 2024 – Apr 2024

Quantitative Developer Intern | Java, Python, Microsoft Azure

Toronto, ON

- Developed an in-house **risk management system**, from raw data processing to deployment on **Microsoft Azure**, saving **\$300,000 annually** by replacing a third-party service
- Built quantitative analysis tools in Python to reveal factor bias and timing patterns from factor models
- Presented monthly insights about portfolio factor risk to **5 portfolio managers** overseeing **\$2 billion AUM**

Bank of America

May 2023 – Aug 2023

Software Developer Intern | Scala, C#

Toronto, ON

- Developed middleware API service in **Scala** to stream real-time payout calculations to a positions dashboard for eSWAP, an equity swaps trading platform processing over **\$40 billion** in transactions annually
- Refactored API routes to decrease the average size of HTTP payloads by **85%** and improve error handling

PROJECTS

Equity Factor Model | Python

2025

- Decomposed historical active returns into Fama-French-Carhart 4 factor model to reduce unwanted factor tilts as a **portfolio manager** on University of Waterloo's Student Investment Fund (\$600k AUM)

Chess Engine | C++

2023

- Chess engine in **C++20** with Negamax search, Alpha-Beta pruning, and piece-square table evaluation

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, Java, C#, Scala, SQL, R

Libraries/Frameworks: NumPy, pandas, polars, Ray, scikit-learn, Flask, Anthropic SDK, React, Node.js, Next.js

Tools: Git, Linux, Docker, Airflow, AWS, Microsoft Azure, Vercel, DataDog, Langfuse, Bloomberg Terminal